INCH-POUND

MIL-M-38510/112A
AMENDMENT 3
09 Feb 1995
SUPERSEDING
AMENDMENT 2
22 March 1987

MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR, VOLTAGE COMPARATORS,
MONOLITHIC SILICON

This amendment forms a part of MIL-M-38510/112A, dated 27 January 1986, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.2.3, Outline letter A, case outline column, delete "F-14" and substitute "F-1".
1.2.3, Case outline, add "2 C-2 (20-terminal SQ.CCP (.350" x .350")".
1.3, Absolute maximum ratings, sink current, delete "29" and substitute "20".

PAGE 2

1.5, Add "2 20-terminal SQ.CCP 90 mW @ T_A = 125°C 55°C 121°C/W".
* 3.1, Detail specifications, delete in its entirety and substitute "3.1 Item requirements. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein. The individual item requirements for devices marked with the "Q" certification mark shall be in accordance with MIL-I-38535 and as specified herein or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not effect the form, fit, or function as described herein."

PAGE 3

* Table I, add new footnote "1/" to the conditions column heading.
Table I, conditions column heading, line 3, delete "figure 4" and substitute "figure 6". Make this same change wherever it occurs within the table.
* Table I, Input bias current, $+I_{IB}$, and $-I_{IB}$, delete "-1" max and substitute "+0.1" max (4 times).
* Table I, Input offset voltage tests, footnote "1/" will be renumbered as footnote "2/".
* Table I, Input offset current tests, footnote "1/" will be renumbered as footnote "2/".
* Table I, Input bias current tests, footnote "1/" will be renumbered as footnote "2/" (2 times).
* Table I, Input voltage common mode rejection tests, footnote "2/" will be renumbered as footnote "3/".

PAGE 4

* Table I, add new footnote "1/" to the conditions column heading.
Table I, t_{RLH}, conditions column, R_L = 5.1 kΩ, add "See figure 7". Make this same addition for t_{RHL} on page 5.

PAGE 5

* Table I, add new footnote "1/" to the conditions column.
* Footnotes, Add new footnote as follows, "1/ For devices marked with the "Q" certification mark, the parameters listed herein may be guaranteed if not tested to the limits specified in accordance with the manufacturer's QM plan."
* Footnote "1/" will be renumbered as footnote "2/".
* Footnote 2/, delete in its entirety and substitute "3/" CMR shall be calculated from V_{IO} measurements defined in note 2/."
* Footnote 1/ table, $-V_{CC}$ column, delete "3 V" and substitute "-3 V".

AMSC N/A
FSC 5962

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202610-21T13:22Z
Check the source to verify that this is the current version before use.

* 4.1, Sampling and inspection, delete in its entirety and substitute "4.1 Sampling and Inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5004, 5005, and 5007, as applicable, of MIL-STD-883, except as modified herein. For devices marked with the "Q" certification mark, sampling and inspection procedures shall be in accordance with MIL-I-38535 or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not effect the form, fit, or function as described herein.

* Figure 2, heading, delete "Device types 01 and 02, (1/4 device type 01, 1/2 device type 02)" and substitute "Device type 01, 2 times device type 02".

* Figure 3, device type 02, delete and substitute the following:

DEVICE TYPE 02

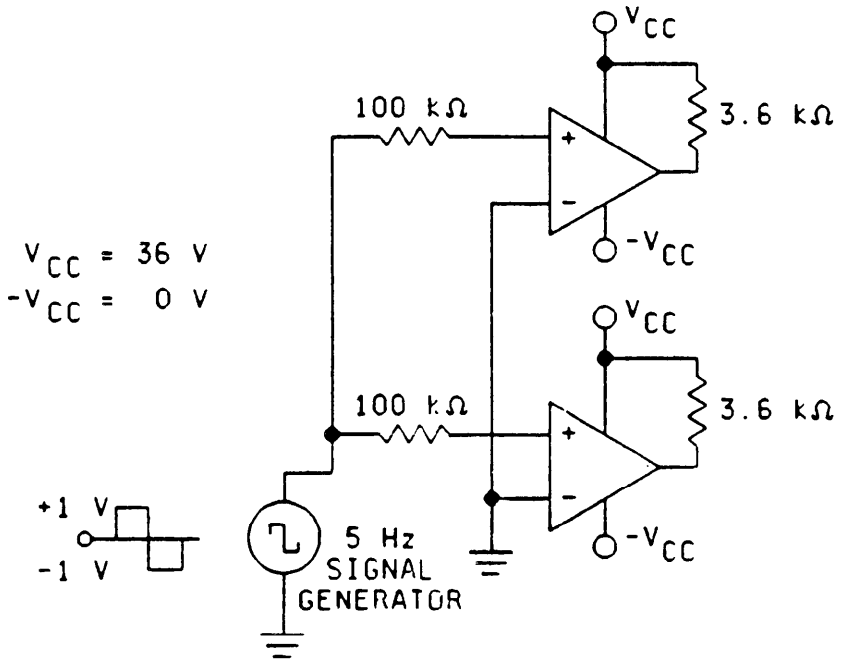


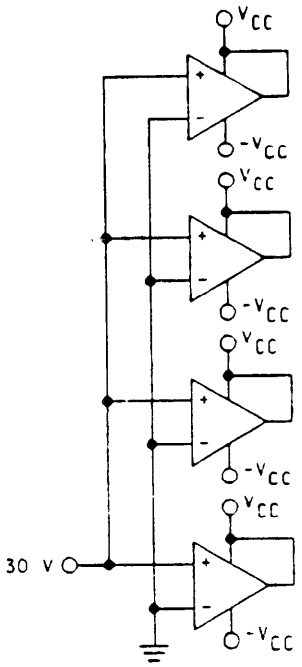
Figure 4, delete "V_{CC} = 0 V" and substitute "$-V_{CC}$ = 0 V".

* Figure 4, device type 02, burn-in test circuit, add a 6.2 kΩ resistor in series with the 30 V input line.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:22Z
Check the source to verify that this is the current version before use.

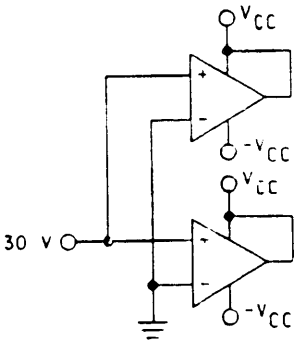
* Figure 5, delete in its entirety and substitute the following:

DEVICE TYPE 01



V_{CC} = 36 V
$-V_{CC}$ = 0 V

DEVICE TYPE 02



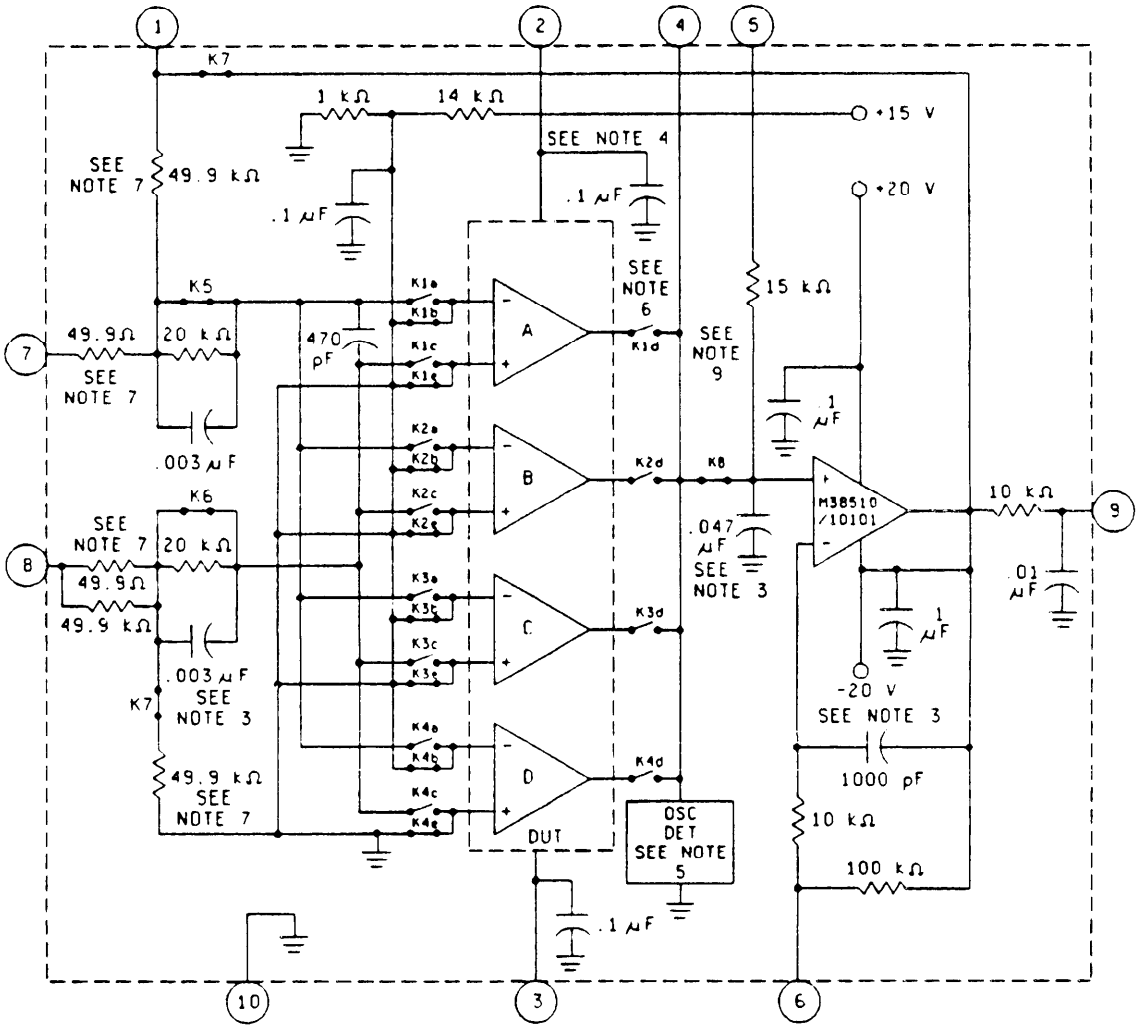
NOTE: If accelerated, high temperature test conditions are used, the device manufacturer shall ensure that at least 85% of the applied voltage is dropped across the device under test at temperature. The device is not considered functional under accelerated test conditions.

FIGURE 5. Accelerated burn-in and life test circuits

Source: https://assist.dla.mil -- Downloaded: 202610-21T13:22Z
Check the source to verify that this is the current version before use.

* Figure 6, delete in its entirety and substitute the following:

DEVICE TYPE 01



See following notes.

FIGURE 6. Test circuit for static and dynamic tests.

Source: https://assist.dla.mil -- Downloaded: 202610-21T13:22Z
Check the source to verify that this is the current version before use.

* Figure 6, delete in its entirety and substitute the following:

DEVICE TYPE 02

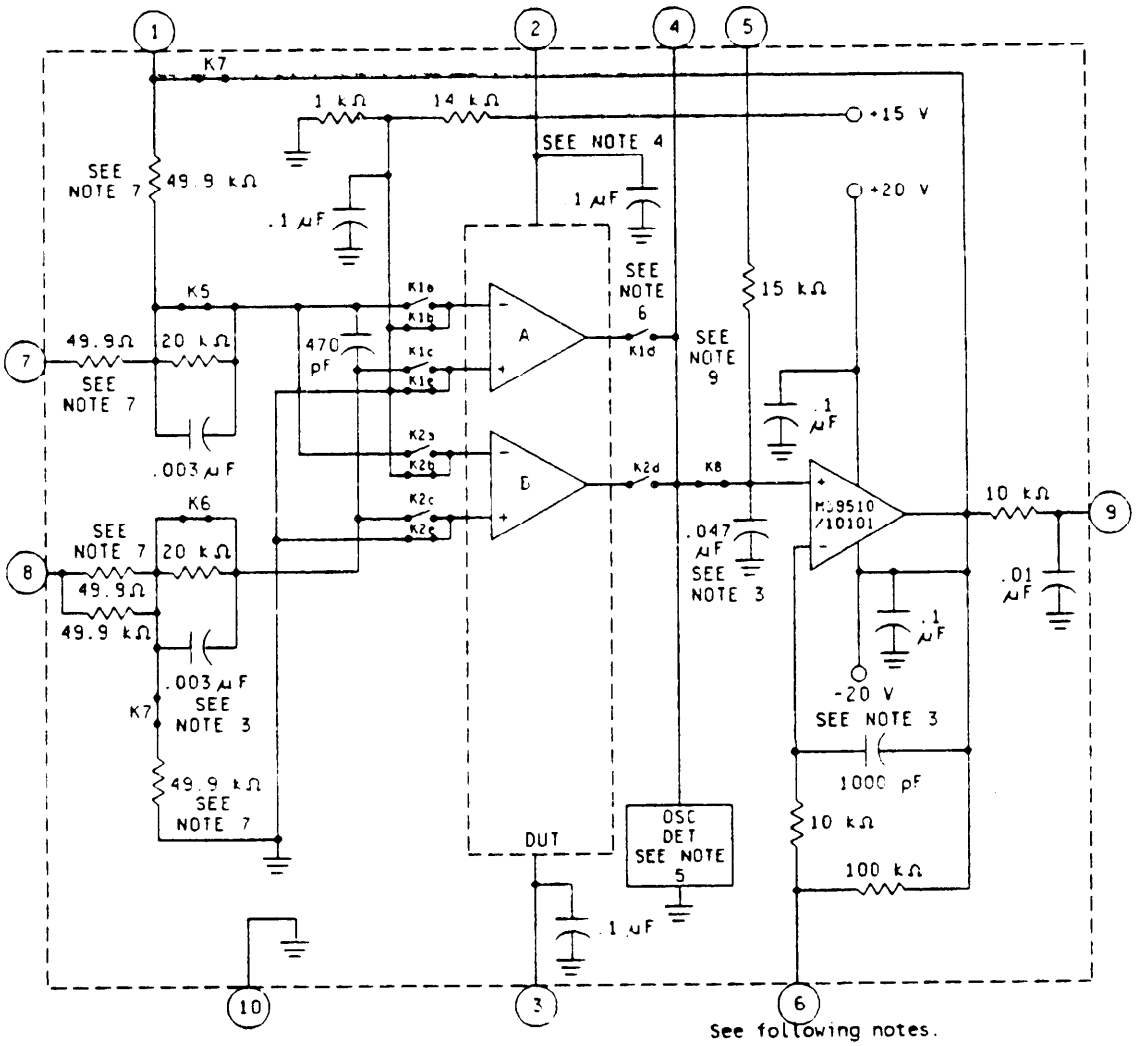


FIGURE 6. Test circuit for static and dynamic tests - Continued.

PAGE 18

* Figure 7, Note 2a, delete in its entirety and substitute "a. With V_{IN} = 0, adjust V_{ref} from -5.0 mV to +5.0 mV for subgroup 1, adjust V_{ref} from -7.0 mV to +5.0 mV for subgroups 2 and 3, in 0.1 mV steps and stop when output switches from high to low."

Source: https://assist.dla.mil -- Downloaded: 202610-21T13:22Z
Check the source to verify that this is the current version before use.

* Figure 8, delete in its entirety and substitute the following:

DEVICE TYPE 01

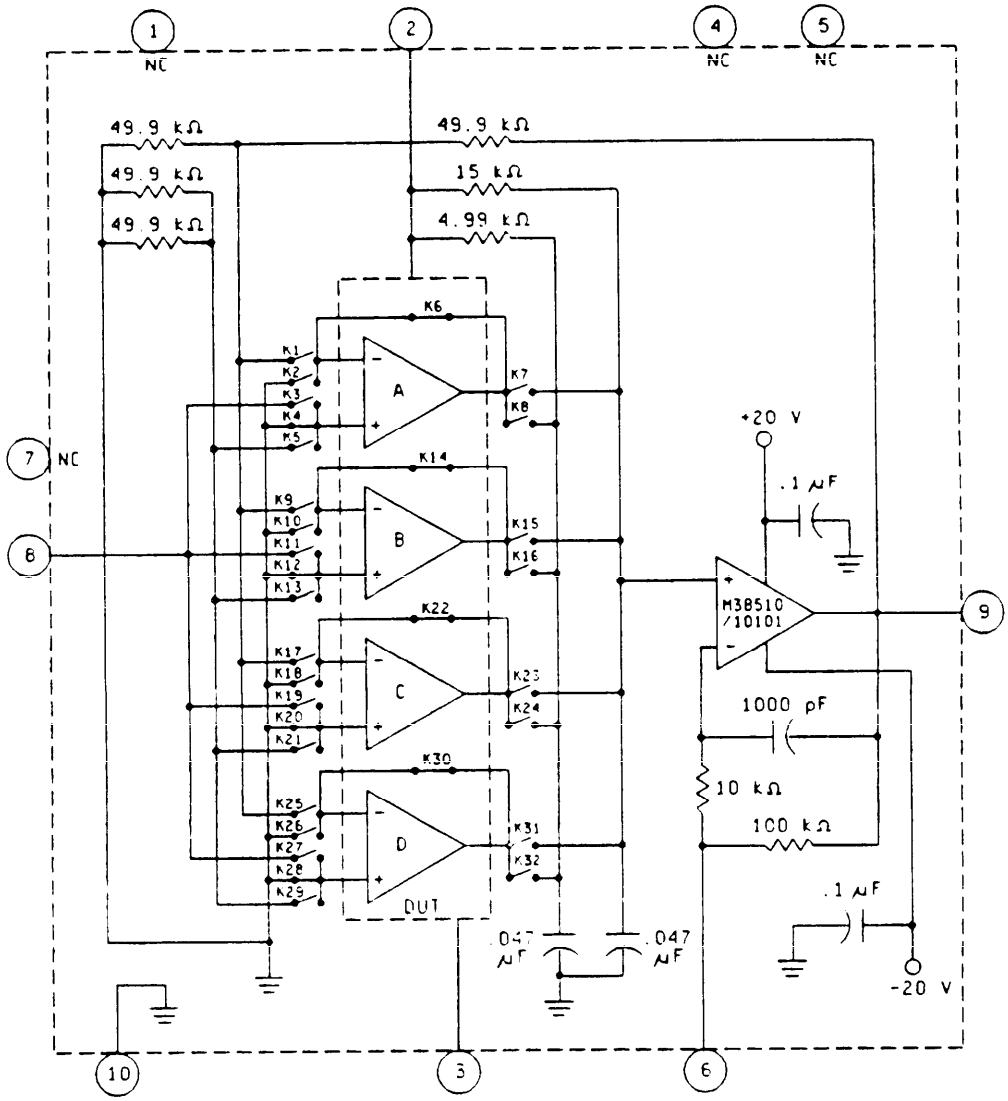


FIGURE 8. Channel separation test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:22Z
Check the source to verify that this is the current version before use.

* Figure 8, delete in its entirety and substitute the following:

DEVICE TYPE 02

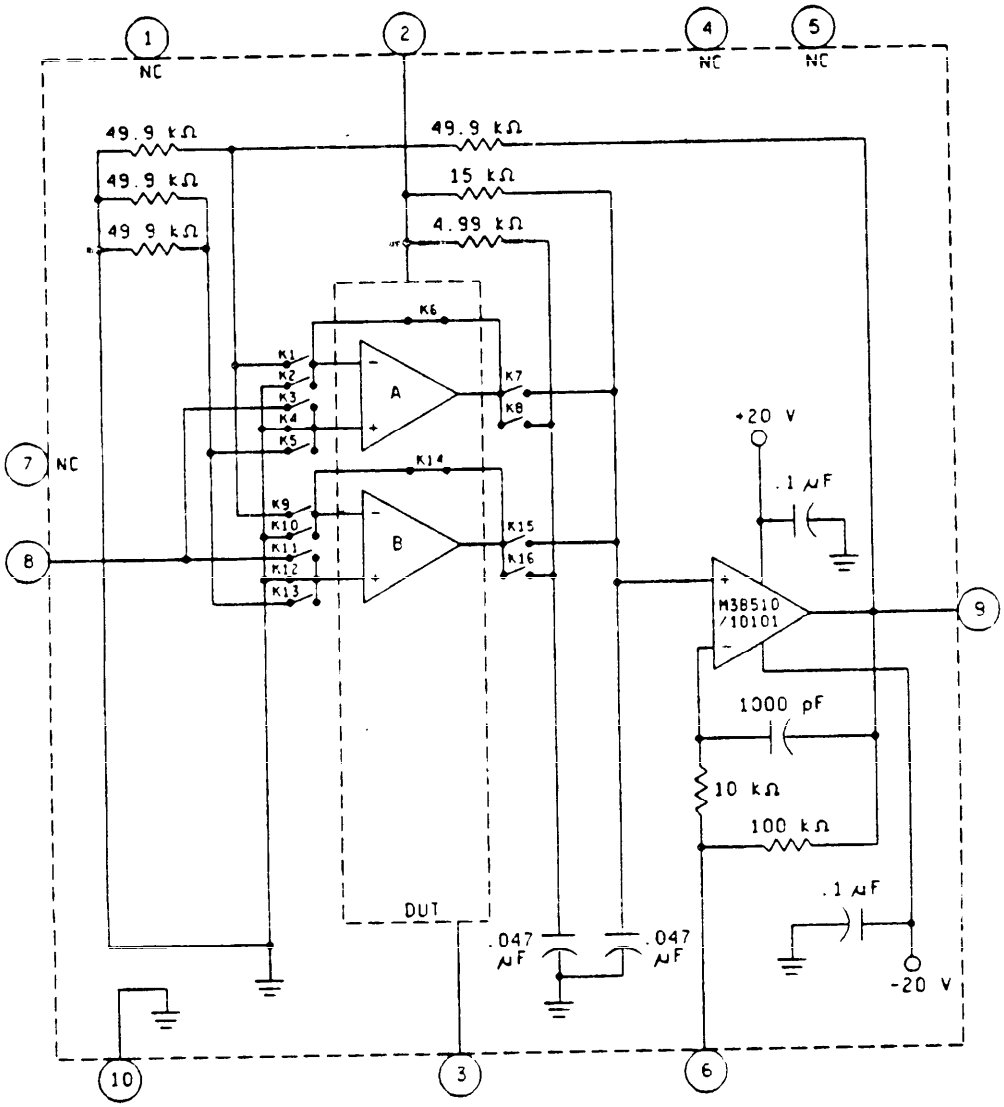


FIGURE 8. Channel separation test circuit - Continued.

* Figure 8, note 4, delete its entirety and substitute "4. The nulling amplifier shall be a JM38510/10101XXX or equivalent. Saturation of the nulling amplifier is not allowed."

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:22Z
Check the source to verify that this is the current version before use.

PAGE 23

* Table III, add new footnote 1/ to the notes heading column.
* Table III, V_{IO}, subgroups 1 and 2, footnote "1/" will be renumbered as footnote "2/" (8 times).
* Table III, V_{IO}, subgroup 1, footnote "11/" will be renumbered as footnote "12/" (4 times).
* Table III, V_{IO}, subgroups 1 and 2, footnote "10/" will be renumbered as footnote "11/" (8 times).
* Table III, I_{IO}, subgroups 1 and 2, footnote "1/" will be renumbered as footnote "2/" (8 times).
* Table III, I_{IO}, subgroups 1 and 2, footnote "10/" will be renumbered as footnote "11/" (8 times).
* Table III, I_{IO}, subgroup 1, footnote "11/" will be renumbered as footnote "12/" (4 times).
* Table III, $+I_{IB}$, subgroups 1 and 2, footnote "1/" will be renumbered as footnote "2/" (8 times).
* Table III, $+I_{IB}$, subgroups 1 and 2, footnote "10/" will be renumbered as footnote "11/" (8 times).
* Table III, $+I_{IB}$, subgroup 1, footnote "11/" will be renumbered as footnote "12/" (4 times).
* Table III, $-I_{IB}$, subgroups 1 and 2, footnote "1/" will be renumbered as footnote "2/" (8 times).
* Table III, $-I_{IB}$, subgroups 1 and 2, footnote "10/" will be renumbered as footnote "11/" (8 times).
* Table III, $-I_{IB}$, subgroup 1, footnote "11/" will be renumbered as footnote "12/" (4 times).
* Table III, CMR, subgroup 1, footnote "2/" will be renumbered as footnote "3/" (2 times).
* Table III, CMR, subgroup 1, footnote "8/" will be renumbered as footnote "9/" (2 times).
* Table III, CMR, test number 18, equation column, delete "CMR = 20 log $|30000/E_3 - E_4|$" and substitute "CMR = 20 log $|3000/E_3 - E_4|$".
* Table III, I_{CEX}, subgroup 1, footnote "3/" will be renumbered as footnote "4/".
* Table III, I_{CEX}, subgroup 1, footnote "10/" will be renumbered as footnote "11/".
* Table III, $+I_{IL}$, subgroup 1, footnote "7/" will be renumbered as footnote "8/".
* Table III, $-I_{IL}$, subgroup 1, footnote "7/" will be renumbered as footnote "8/".
* Table III, $+I_{IL}$, subgroup 1, footnote "10/" will be renumbered as footnote "11/".
* Table III, $-I_{IL}$, subgroup 1, footnote "10/" will be renumbered as footnote "11/".
* Table III, V_{OL}, subgroup 1, footnote "10/" will be renumbered as footnote "11/".
* Table III, I_{CC}, subgroup 1, footnote "4/" will be renumbered as footnote "5/".
* Table III, $\Delta V_{IO}/\Delta T$, subgroup 2, footnote "6/" will be renumbered as footnote "7/".
* Table III, $\Delta I_{IO}/\Delta T$, subgroup 2, footnote "6/" will be renumbered as footnote "7/".

PAGE 24

* Table III, add new footnote 1/ to the notes heading column.
* Table III, CMR, subgroups 2 and 3, footnote "2/" will be renumbered as footnote "3/" (4 times).
* Table III, CMR, subgroups 2 and 3, footnote "8/" will be renumbered as footnote "9/" (4 times).
* Table III, CMR, test number 45, equation column, delete "CMR = 20 log $|30000/E_{21} - E_{22}|$" and substitute "CMR = 20 log $|3000/E_{21} - E_{22}|$".
* Table III, I_{CEX}, subgroups 2 and 3, footnote "3/" will be renumbered as footnote "4/" (2 times).
* Table III, I_{CEX}, subgroups 2 and 3, footnote "10/" will be renumbered as footnote "11/" (2 times).
* Table III, $+I_{IL}$, subgroups 2 and 3, footnote "7/" will be renumbered as footnote "8/" (2 times).
* Table III, $-I_{IL}$, subgroups 2 and 3, footnote "7/" will be renumbered as footnote "8/" (2 times).
* Table III, $+I_{IL}$, subgroups 2 and 3, footnote "10/" will be renumbered as footnote "11/" (2 times).
* Table III, $-I_{IL}$, subgroups 2 and 3, footnote "10/" will be renumbered as footnote "11/" (2 times).
* Table III, V_{OL}, subgroups 2 and 3, footnote "10/" will be renumbered as footnote "11/" (4 times).
* Table III, I_{CC}, subgroups 2 and 3, footnote "4/" will be renumbered as footnote "5/" (4 times).
* Table III, V_{IO}, subgroup 3, footnote "1/" will be renumbered as footnote "2/" (4 times).
* Table III, V_{IO}, subgroup 3, footnote "10/" will be renumbered as footnote "11/" (4 times).
* Table III, $\Delta V_{IO}/\Delta T$, subgroup 3, footnote "6/" will be renumbered as footnote "7/".
* Table III, I_{IO}, subgroup 3, footnote "1/" will be renumbered as footnote "2/" (4 times).
* Table III, I_{IO}, subgroup 3, footnote "10/" will be renumbered as footnote "11/" (4 times).
* Table III, $\Delta I_{IO}/\Delta T$, subgroup 3, footnote "6/" will be renumbered as footnote "7/".
* Table III, $+I_{IB}$, subgroup 3, footnote "1/" will be renumbered as footnote "2/" (4 times).
* Table III, $+I_{IB}$, subgroup 3, footnote "10/" will be renumbered as footnote "11/" (4 times).
* Table III, $-I_{IB}$, subgroup 3, footnote "1/" will be renumbered as footnote "2/" (4 times).
* Table III, $-I_{IB}$, subgroup 3, footnote "10/" will be renumbered as footnote "11/" (4 times).
* Table III, CMR, test number 72, equation column, delete "CMR = 20 log $|30000/E_{39} - E_{40}|$" and substitute "CMR = 20 log $|3000/E_{39} - E_{40}|$".

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:22Z
Check the source to verify that this is the current version before use.

PAGE 25

* Table III, add new footnote 1/ to the notes heading column.
* Table III, A_{VS}, subgroups 4, 5, 6, footnote "5/" will be renumbered as footnote "6/" .
* Table III, A_{VS}, subgroups 4, 5, 6, footnote "10/" will be renumbered as footnote "11/" .

PAGE 26

* Table III, add new footnote 1/ to the notes heading column.
* Footnotes, Add new footnote as follows, "1/ For devices marked with the "Q" certification mark, the parameters listed herein may be guaranteed if not tested to the limits specified in accordance with the manufacturer's QM plan."
* Footnote "1/" will be renumbered as footnote "2/".
* Footnote "2/" will be renumbered as footnote "3/".
* Footnote "3/" will be renumbered as footnote "4/".
* Footnote "4/" will be renumbered as footnote "5/".
* Footnote "5/" will be renumbered as footnote "6/".
* Footnote "6/" will be renumbered as footnote "7/".
* Footnote "7/" will be renumbered as footnote "8/".
* Footnote "8/" will be renumbered as footnote "9/".
* Footnote "9/" will be renumbered as footnote "10/".
* Footnote "10/" will be renumbered as footnote "11/".
* Footnote "11/" will be renumbered as footnote "12/".

PAGE 27

4.4.3b, delete and substitute:
"b. Steady-state life test for class B (method 1005 of MIL-STD-883).
Test condition D using the circuit shown on figure 3, test condition C using the circuit shown on figure 4, or test condition F using the circuit shown on figure 5. (See 4.3a.2, note.)"
4.4.3c, delete in its entirety.
4.5.2, line 3, between condition and prior, add "at room temperature".

PAGE 28

6.2g, delete "carriers,". Delete the last sentence in its entirety.

The margins of this amendment are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) to the previous amendment were made. This was done as convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
7	Reprinted without change
8	8

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:22Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI, SM
Navy - SH, OS, TD, MC, CG, AS
Air Force - 19, 85, 99
DLA - ES

Civil Agency Coordinating Activity:
NASA-NA

Preparing activity:
DLA-ES

Agent:
DLA - ES

(Project 5962-1405)

Source: https://assist.dla.mil -- Downloaded: 202610-21T13:22Z
Check the source to verify that this is the current version before use.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.3 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in (method 1015 of MIL-STD-883).

1. For class S devices: Test condition D using the circuit shown on figure 3.

2. For class B devices: Test condition D using the circuit shown on figure 3, test condition C using the circuit shown on figure 4, or test condition F using the circuit shown on figure 5.

NOTE: If accelerated high-temperature test conditions are used, the device manufacturer shall ensure that at least 85 percent of the applied voltage is dropped across the device at temperature. The device is not considered functional under accelerated test conditions.

b. Reverse bias burn-in (method 1015 of MIL-STD-883). For class S devices only using the circuit shown on figure 4.

c. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

d. The percent defective allowable (PDA) - The PDA for class S and class B devices shall be as specified in MIL-M-38510, based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests are performed prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in divided by the total number of devices submitted for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Subgroups 10 and 11 shall be omitted.

b. Electrical tests requirements shall be as specified in table II herein.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady-state life test for class S (method 1005 of MIL-STD-883). Test condition F using the circuit shown on figure 5 (see 4 3a.2 note). If the alternate burn-in conditions are used, the circuit shown on figure 4 shall be used.

Source: https://assist.dla.mil -- Downloaded: 202610-21T13:22Z
Check the source to verify that this is the current version before use.

DEVICE TYPE 01

CASE 2

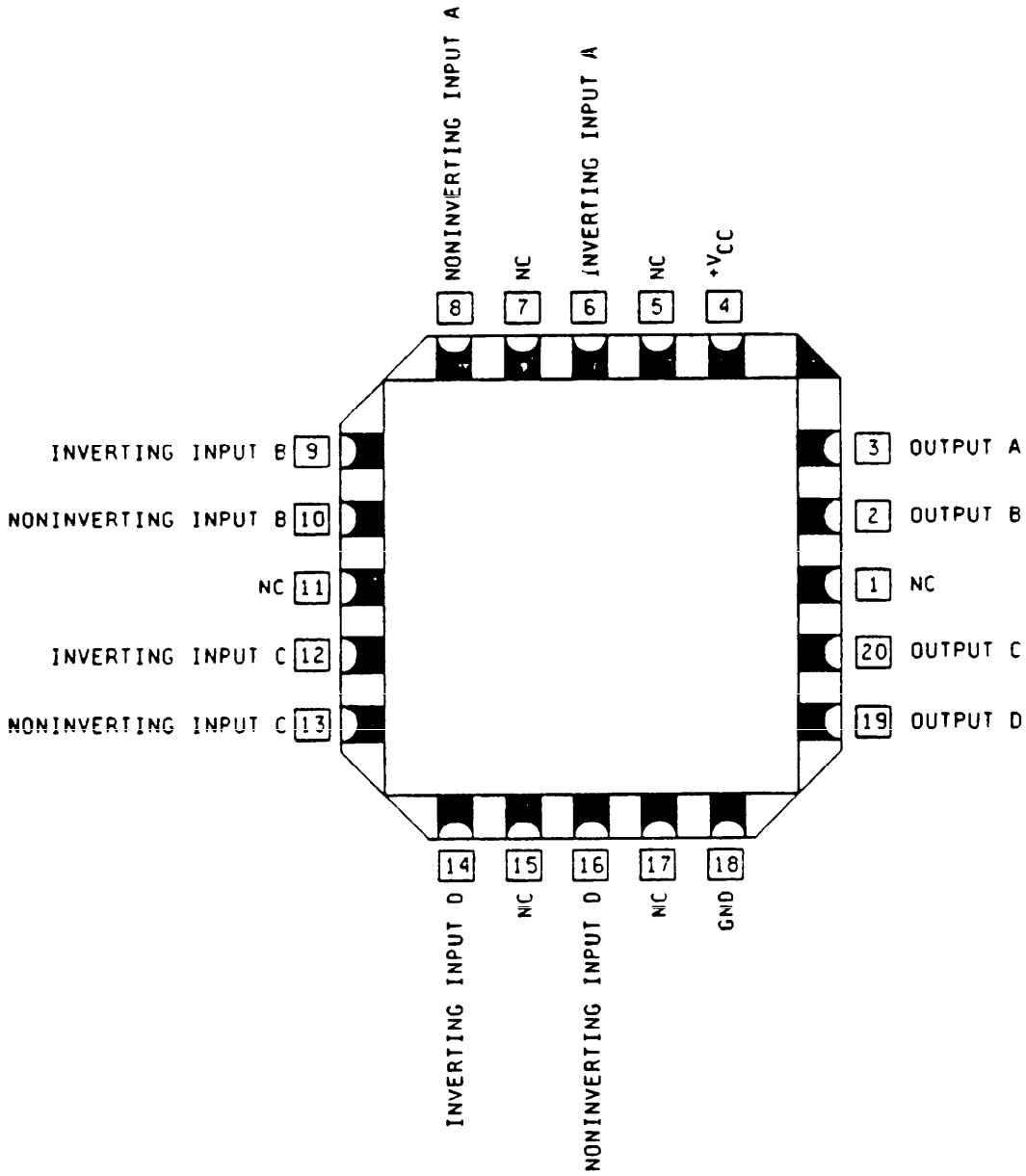


FIGURE 1. Terminal connections.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:22Z
Check the source to verify that this is the current version before use.